SIX WEEKS INDUSTRIAL TRAINING PROJECT REPORT

ON

“CRYPTO-TRACKER”

Submitted in the partial fulfilment of the requirement for the Award of Degree of

Bachelor of Technology

In

Computer Science & Engineering

Session: 2023-2027

SUBMITTED TO:

Dr. Sandip Goyal

Head of Department

SUBMITTED BY:

Pari Agarwal

11232757

Department of Computer Science & Engineering

MAHARISHI MAHAKANDESHWAR DEEMED TO BE UNIVERSITY, MULLANA

CERTIFICATE BY COMPANY

Solitaire Infosys Simple Creative innovative

S.No. 544682

Certificate of Training

This certificate has been awarded to Ms **Pari Agarwal** from **MMDU, Mullana** who has undertaken an internship program of **6 Weeks** from **05/05/2025** to **18/06/2025** in **React JS** Department from Solitaire Infosys Pvt. Ltd.

During the tenure of this internship with us, we found the candidate self-starter and hardworking. Also she had worked sincerely on the assignments and her performance was satisfactory to be part of the team.

We wish the Candidate success for all the future endeavors.

For Solitaire Infosys Pvt. Ltd.

CERTIFIED ISO 9001 2015 COMPANY

Human Resources Department

Note: To check the authentication of certificate, please visit www.slnfy.com

DECLARATION BY STUDENT

I hereby declare that the Industrial Training Report entitled "CRYPTO-TRACKER" is an authentic record of my own work carried out at **Solitaire Infosys Pvt. Ltd.** as requirement of 6-weeks Industrial Training during the period from May 2025 to June 2025 for the award of degree of **B.Tech. CSE, Maharishi Mahakandeshwar (Deemed to be University), Mullana** under the guidance of **Mr. Sahil Sandhu.**

PARI AGARWAL
(11232757)

Date: _____

Certified that the above statement made by the student is correct to the best of our knowledge and belief.

Industrial Supervisor
(Signature and Seal)

ACKNOWLEDGEMENT

Through this acknowledgement, I express our sincere gratitude to all those people who have been associated with this project and have helped me with it and made it a worthwhile experience. First and foremost, I would like to thank Solitaire Infosys for giving me courage to make this project. At the outset, I would like to propose a word of thanks for the people who gave me unending support and help in numerous ways.

Firstly, I express our thanks to **Dr. Sandip Goyal** (H.O.D CSE DEPT.) who gave me this opportunity to learn the subject in a practical approach who guided me and gave me valuable suggestions regarding the project report.

Secondly, I would like to thanks **Mr. Sahil Sandhu** (Team Leader) and FACULTY MEMBERS of Solitaire Infosys who gave me their immense support in completing the project. The atmosphere provided was full of gaining more and more knowledge and to keep enthusiastic nature. The teachers provided a lot of help in resolving our doubts and making this project successful in shorter time.

Lastly, I would also like to thank my parents and friends who helped us a lot in finishing the project in limited time.

I make this project not only for marks but also to increase our knowledge. Thanks again to all who helped me in completion of this project.

Pari Agarwal

5th Sem

B.tech (Computer Science & Engg.)

ABOUT COMPANY

Solitaire Infosys Pvt. Ltd. is an acclaimed IT service provider contributing its part in the development of many businesses around the globe. We socialize with our clients to get a superior cognizance of their business and requirements and help them in fabricating websites and applications for their business. Founded in 2011 by a dynamic duo with the same aim and zeal, we have come a long way in satisfying our clients.

We are serving our clients with the world-class services for more than seven years now. The clients are delivered with the best IT solutions after we have developed a great understanding of their business and requirements. Our team works on the client projects like its own and that is the reason why we hold the edge in the league.

With every project that we deliver, we deliver our respect, creativity, quality, transparency, and teamwork to our clients. We have the experience, expertise, and capabilities to enable organizations to accelerate their service processes in every possible way. We are known for our excellent customer satisfaction, cost-effectiveness, and innovative skills that are unparalleled.

Services :

1. Mobile Applications

2. Web Development

3. Digital Marketing

4. UX & UI Design

5. Enterprise Applications

TABLE OF CONTENT

S.NO	TOPIC	PAGE NO.
1.	**Introduction** 1.1 Objective 1.2 Scope of project 1.3 Existing System 1.4 Proposed System	8 - 9
2.	**Feasibility Study** 2.1 Operation Feasibility 2.2 Technical Feasibility 2.3 Economical Feasibility	10 - 11
3.	**Tools & Technology Used** 3.1 Hardware Requirements 3.2 Software Requirements 3.3 Technology used: 1. HTML 2. CSS 3. JAVASCRIPT 4. BOOTSTRAP 5. REACT	12 - 19
4.	**Data Flow Diagram**	20
5.	**Screenshots of Project**	21
6.	**Conclusion and Future Scope** 6.1 Conclusions	26

	6.2 Further and Future Scope	
7.	**References**	27

CHAPTER - 1

INTRODUCTION

Crypto-Tracker is a web application entirely focused on the cryptocurrency industry, offering an online centralized multi-function crypto tracking application. The vision behind our thought is to provide every cryptocurrency detail that users want at their fingertips, along with real-time analysis and secure transaction management. The main vision in building this web application is to empower crypto enthusiasts, traders, and investors with accurate data and efficient tools that enhance their decision-making in the rapidly evolving world of digital currencies.

Crypto-Tracker is a web application. An online centralized multi-function cryptocurrency application.

The main objective of this web application is to make cryptocurrency tracking and management easier and more accessible. A fully-featured platform that consolidates live coin tracking, price comparisons, historical data visualization, and user-friendly features like wishlisting and email sharing—all available to users 24×7. It also aims at promoting digital literacy and secure management of crypto data.

This project aims to raise awareness about using technology for smarter crypto investing and managing digital assets responsibly. We are planning to build a community that takes care of crypto investors, enthusiasts, and traders, supporting them with accurate data, intuitive tools, and a seamless experience. Let's start with ourselves and become the change we want to see in the crypto world.

That's why we believe in:

"TRACK THE TREND, BEFRIEND THE TREND."

1.1 Objectives

1. To provide every cryptocurrency detail that users want, at their fingertips.
2. To enable secure, real-time comparisons and price tracking of cryptocurrencies.
3. To empower users with data-driven insights for informed decision-making.
4. To make crypto investing and tracking easier and more reliable.

1.2 Scope:

The Crypto-Tracker web application aims to serve as an all-in-one platform for crypto investors, traders, and enthusiasts. It provides real-time cryptocurrency prices, historical data analysis, price comparisons, wishlisting, email sharing, and secure user authentication. The platform is designed to be user-friendly, scalable, and accessible from anywhere, at any time.

1.3 Existing system

Currently, many crypto enthusiasts and traders depend on multiple fragmented platforms to monitor, analyze, and compare cryptocurrency data. This often results in a disjointed experience, as users have to switch between different apps and websites to get information like price updates, chart visualizations, and analytics. Most of these platforms lack personalized features such as wishlisting or tailored alerts, leaving users without an easy way to organize and prioritize their interests. Additionally, sharing insights or specific coin data with others typically requires manual steps, like screenshots or using third-party apps, which can be time-consuming and inconvenient

1.4 Proposed system

The proposed system, Crypto-Tracker, aims to solve these limitations by offering a centralized and feature-rich platform tailored to the needs of crypto investors, enthusiasts, and traders. This system consolidates real-time prices, historical data, interactive charts, and personalized watchlists into a single, easy-to-navigate dashboard. It allows users to create wishlists, receive customized alerts, and quickly access the information they need to make informed decisions.

CHAPTER - 2

FEASIBILITY STUDY

Generally, feasibility studies precede technical development and project implementation. Traditionally uncover the strengths and weaknesses of the existing system or proposed venture. In its simplest term, the two criteria to judge feasibility are cost required and value to be attained. As such, a well-designed feasibility study should provide historical background of the project assessment of feasibility study is based on the following factors:

1) Technical Feasibility
2) Economic Feasibility
3) Operational Feasibility

2.1 Technical Feasibility:

Generally, feasibility studies precede technical development and project implementation. The assessment for **CryptoTracker** is based on system requirements in terms of input (user data, coin data from APIs), processes (fetching, processing, analyzing data), output (charts, tables, notifications), and interfaces (UI/UX design). This includes data volume, trends, and frequency of updates, helping estimate whether the system will perform adequately or not. Technological feasibility is carried out to determine the capability, in terms of software (React.js, Firebase, Chart.js), hardware (modern web servers), personnel (developers, testers), and expertise, to handle the completion of the project.

When writing the feasibility report, the following considerations were made:

- A brief description of the business domain (crypto investment and tracking) and the specific functionality being examined (real-time price tracking, watchlists, comparison, chart visualization, notifications).
- The human and economic factors, including the need for secure authentication and real-time updates to ensure user trust and usability.
- The possible solutions to problems like data accuracy, secure data storage, and user-friendly interfaces.

2.2 Economic Feasibility

Economic analysis is the most frequently used method for evaluating the effectiveness of a new system. More commonly known as cost/benefit analysis, the procedure determines the benefits and savings expected from the candidate system and compares them with the costs. For **Crypto-Tracker**, the benefits include:

- Increased user engagement and retention through easy access to crypto data.
- Time and cost savings by reducing the need to switch between different platforms.
- Improved security and user trust, reducing potential losses from fraud or hacking.

2.3 Operational Feasibility:

Operational feasibility is a measure of how well the proposed system solves the identified problems and meets user needs as outlined during the scope definition and requirements analysis phase. **Crypto-Tracker** aims to solve the problem of fragmented crypto data tracking and provide users with a seamless, all-in-one platform to manage their investments. The system's features—such as real-time price tracking, watchlists, coin comparisons, and intuitive charting—help reduce user frustration and support informed decision-making.

2.4 Conclusion of Feasibility Study:

1. Technical Feasibility: The system can be implemented using modern web development frameworks (React.js), cloud services (Firebase), and charting libraries (Chart.js).
2. Economic Feasibility: The system implementation is cost-effective, particularly for small to medium crypto communities and enthusiasts who can benefit from a unified platform without the need for expensive licenses or hardware.
3. Operational Feasibility: The system efficiently operates, reduces manual efforts in managing and comparing crypto investments, and enhances the overall user experience, reducing time spent on fragmented research and improving decision-making.

CHAPTER - 3

TOOLS & TECHNOLOGIES USED:

3.1 Hardware Requirements

- Processor :- Intel ® Celeron ® CPU N3050
- RAM :- 4 GB
- Hard Disk :- 1TB

3.2 Software Requirements

- Operating system :- Window 10
- Language /front end :-HTML, CSS, JAVASCRIPT, BOOTSTRAP, REACT
- Back End :- Firebase

3.3 Technologies Used:

The **Crypto-Tracker** project leverages a powerful combination of modern **client-side web technologies** and **cloud-based backend services** to deliver a robust, responsive, and user-friendly application. On the front end, it utilizes **HTML**, **CSS**, **JavaScript**, **Bootstrap**, and **ReactJS** to create a rich and interactive user interface that is both visually appealing and highly functional. React's component-based architecture ensures the application is modular, reusable, and scalable, while Bootstrap enhances the design with a clean and mobile-friendly layout.

On the backend, the project integrates **Firebase**, a cloud-based platform developed by Google, which eliminates the need for managing traditional server infrastructure. Firebase provides essential backend services such as **user authentication**, **real-time cloud storage**, and **secure database management** through its **Authentication** and **Cloud Firestore** modules. These services enable seamless login and registration functionality, persistent data storage for user grades and quiz scores, and real-time synchronization of data across devices.

i. FRONTEND:

By combining these technologies, the CryptoTracker application achieves a high level of efficiency, security, and user engagement. It ensures that users can manage their cryptocurrency portfolio effectively, view detailed price charts, compare different coins, monitor real-time market trends, and even receive email alerts — all within a single integrated platform that works seamlessly across desktops, tablets, and smartphones.

1. HTML:

HTML, which stands for Hyper Text Mark up Language . It provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, lists etc as well as for links, quotes, and other items.

- HTML stands for Hyper Text Markup Language
- It is the standard markup language for creating Web pages
- It describes the structure of a Web page
- HTML elements tell the browser how to display the content
- HTML elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.

2.CSS:

- CSS stands for Cascading Style Sheets
- CSS describes how HTML elements are to be displayed on screen, paper, or in other media.
- CSS saves a lot of work. It can control the layout of multiple web pages all at once.
- External stylesheets are stored in CSS files.

There are three ways of inserting a style sheet:

1) Inline CSS
2) Internal CSS
3) External CSS

1). Inline CSS:

Inline CSS contains the CSS property in the body section attached with element is known as inline CSS.

- Inline by using the style attribute inside HTML elements

2). Internal CSS:

This can be used when a single HTML document must be styled uniquely. The CSS rule set should be within the HTML file in the head section i.e. the CSS is embedded within the HTML file.

- Internal by using a <style> element in the <head> section.

3). External CSS:

External CSS contains separate CSS file which contains only style property with the help of tag attributes (For example class, id, heading, etc.).

- External by using a <link> element to link to an external CSS file.

3. JAVASCRIPT:

- JavaScript is the most popular **programming language** in the world and that makes it a programmer's great choice. Once you learnt JavaScript, it helps you developing great front-end as well as back-end software using different JavaScript based frameworks like jQuery, Node.JS etc.

- JavaScript is everywhere, it comes installed on every modern web browser and so to learn JavaScript you really do not need any special environment setup. For example Chrome, Mozilla Firefox, Safari and every browser you know as of today, supports JavaScript.

What can we build using JavaScript?

JavaScript is a widely-used programming language. Given below are some domains/products that can be built using JavaScript:

- **Websites:** JavaScript helps us to add behaviour of our website. It helps users to interact with the website. For e.g. clicking on buttons, saving details, uploading details on the website, etc.

- **Web Servers:** We can make robust server applications using JavaScript. To be precise we use JavaScript frameworks like Node.js and Express.js to build these servers.

- **Game Development:** In Game Development industry, JavaScript is used widely. With the addition of HTML5 Canvas, it's now possible to make 2D and 3D games in JavaScript very efficiently.

Why to learn JavaScript?

JavaScript is the most popular and hence the most loved language around the globe. Apart from this, there are abundant reasons to learn it. Below are a listing of few important points:

- **No need of compilers:**

Since JavaScript is an interpreted language, therefore it does not need any compiler for compilations.

- **Used both Client and Server-side:**

Earlier JavaScript was used to build client-side applications only, but with the evolution of its frameworks namely Node.js and Express.js, it is now widely used for building server-side applications too.

- **Helps to build a complete solution:**

As we saw, JavaScript is widely used in both client and server-side applications, therefore it helps us to build an end-to-end solution to a given problem.

- **Used everywhere:**

JavaScript is so loved because it can be used anywhere. It can be used to develop websites, games or mobile apps, etc.

- **Huge community support:**

JavaScript has a huge community of users and mentors who love this language and take it's legacy forward.

Use Cases in Crypto-tracker:

- Interactive Dashboard.
- Calculating profit /loss.
- Updating state dynamically in React.

4. BOOTSTRAP:

- Bootstrap is a responsive front-end framework that speeds up UI development.
- Provides pre-designed UI components like buttons, navbars, modals, and forms.
- Built on a grid system to support responsive layouts.
- Enhances UI consistency across different devices.

5. REACT: Front-End Library:

React is a component-based JavaScript library used for building dynamic user interfaces.

- **JSX**: A syntax extension combining HTML and JavaScript.

- **Components**: Reusable and independent pieces of UI.

- **Virtual DOM**: Increases performance by minimizing direct DOM manipulation.

- Enables fast development of interactive dashboards and single-page applications (SPAs)

Why to use React?

- **Virtual DOM** –

A virtual DOM object is a representation of a DOM object.

Virtual DOM is actually a copy of the original DOM. Any modification in the web application causes the entire UI to re-render the virtual DOM.

Then the difference between the original DOM and this virtual DOM is compared and the changes are made accordingly to the original DOM.

- **JSX –**

Stands for JavaScript XML. It is an HTML/XML JavaScript Extension which is used in React. Makes it easier and simpler to write React components.

- **Components –**

React JS supports Components. Components are the building blocks of UI wherein each component has a logic and contributes to the overall UI. These

components also promote code reusability and make the overall web application easier to understand.

- **High Performance –**

Features like Virtual DOM, JSX and Components makes it much faster than the rest of the frameworks out there.

- **Developing Android/IOS Apps –**

With React Native you can easily code Android-based or IOS-Based apps with just the knowledge of JavaScript and ReactJS.

- You can start your react application by first installing “create-react-app” using npm or yarn.

ii. BACKEND TECHNOLOGIES

1. Firebase (by Google)

Firebase is a comprehensive Backend-as-a-Service (BaaS) that handles authentication, database, and hosting.

Features Used:

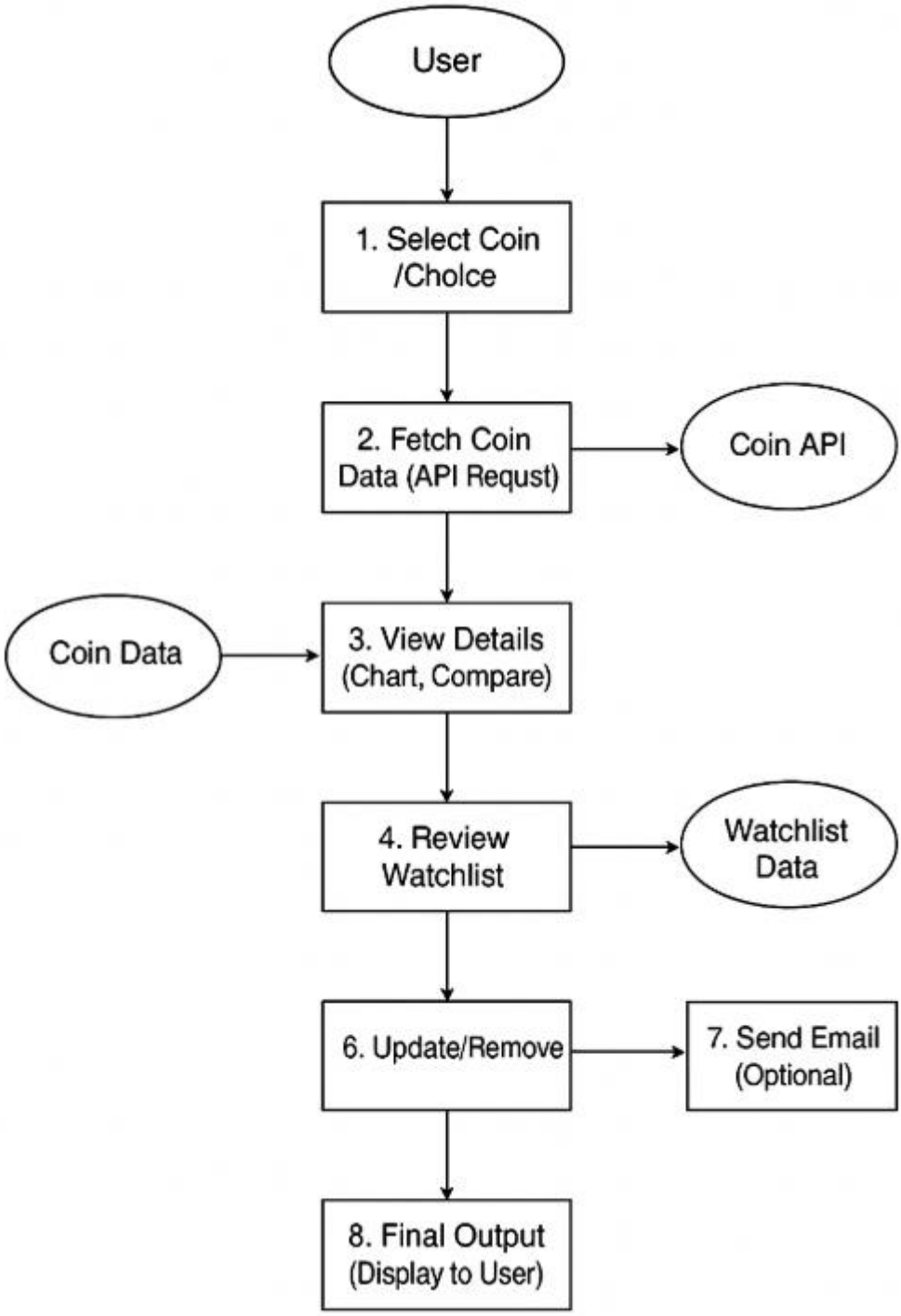
- Firebase Authentication: Used for user sign-up, login, and session management.
- Cloud Firestore: A NoSQL database used to store grade records, quiz scores, and user data in real-time.
- Firebase Hosting: Used to deploy and host the Grade Tracker app securely.

Why Firebase?

- No need for a custom server or database like Node.js or MongoDB.
- Offers secure and real-time backend services.
- Easily integrates with React through Firebase SDKs.
- Automatically scales with application demand.

CHAPTER - 4

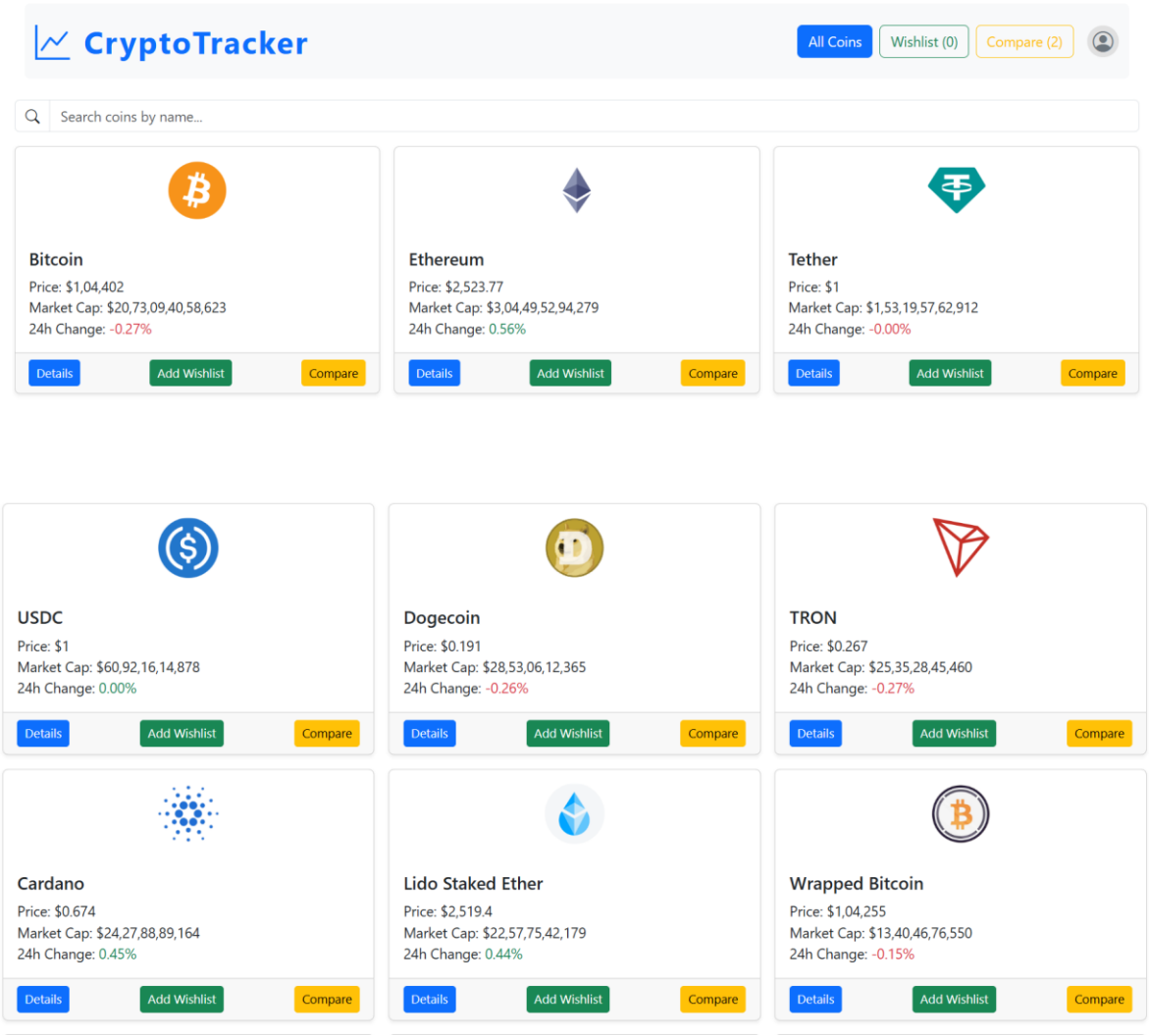
DATA FLOW DIAGRAM



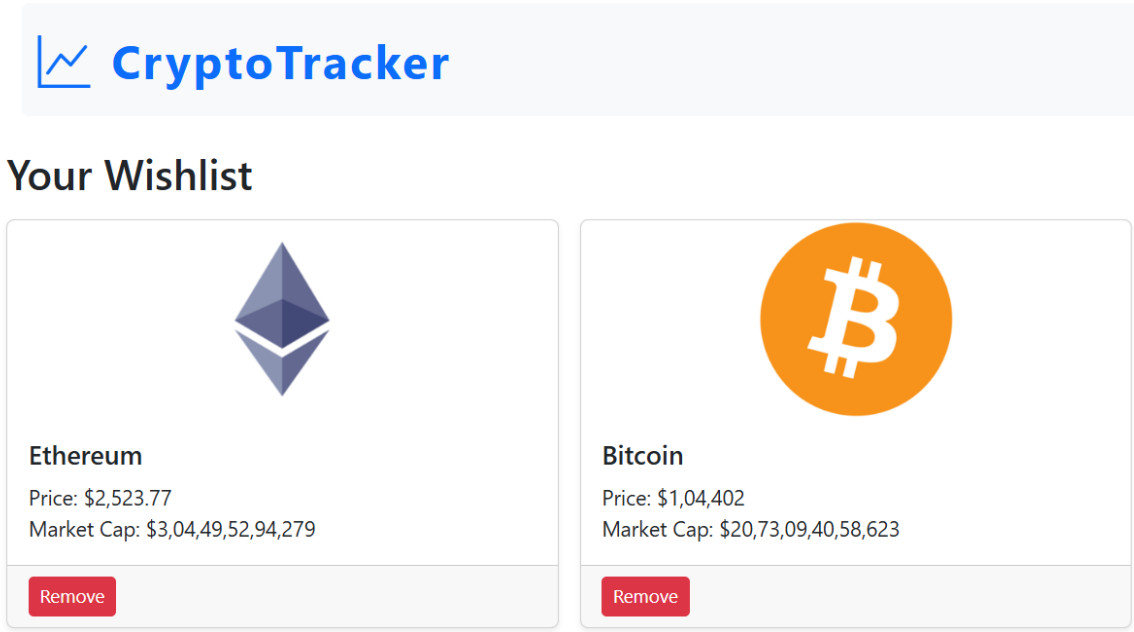
CHAPTER - 5

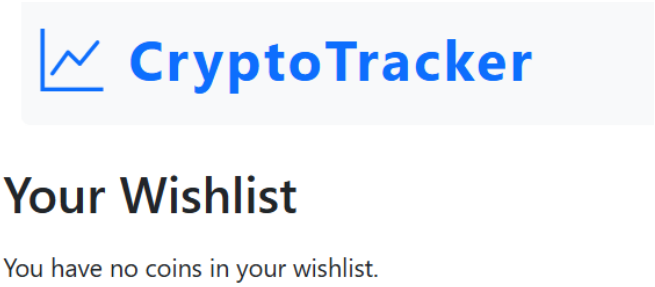
SCREENSHOTS OF PROJECT

HOME:

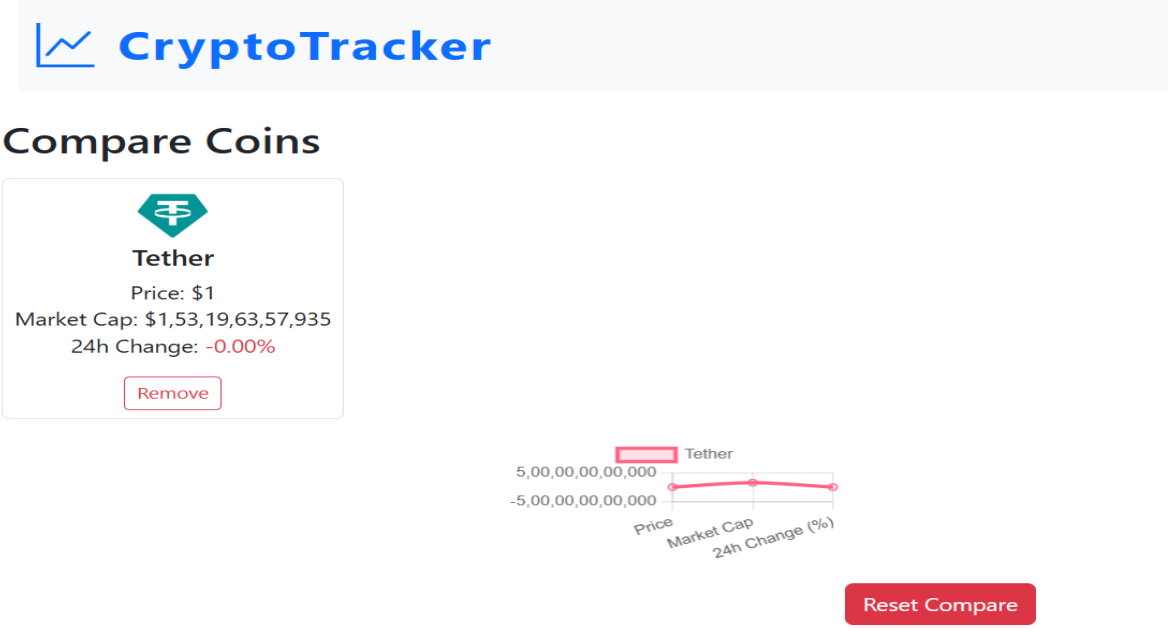


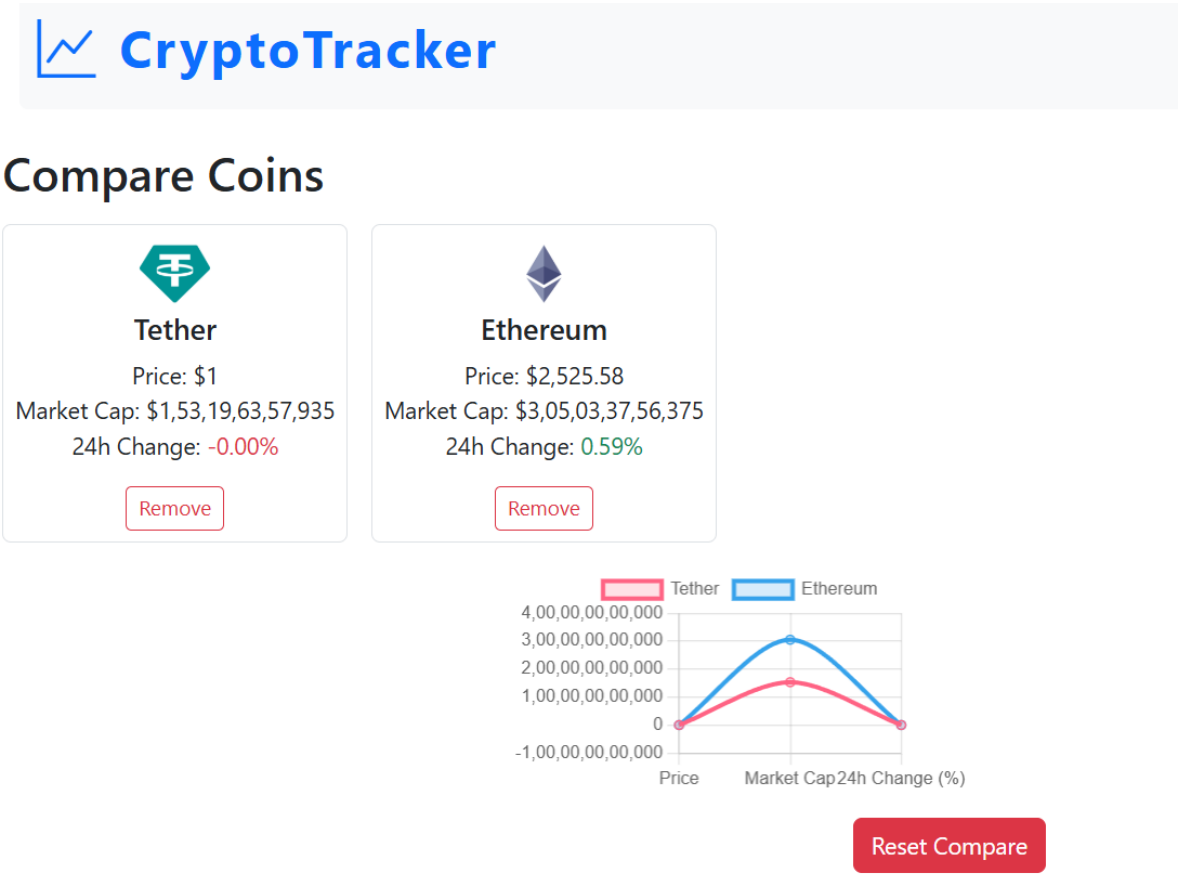
WISHLIST:


CryptoTracker
Your Wishlist
Ethereum
Price: $2,523.77
Market Cap: $3,04,49,52,94,279
Remove
Bitcoin
Price: $1,04,402
Market Cap: $20,73,09,40,58,623
Remove

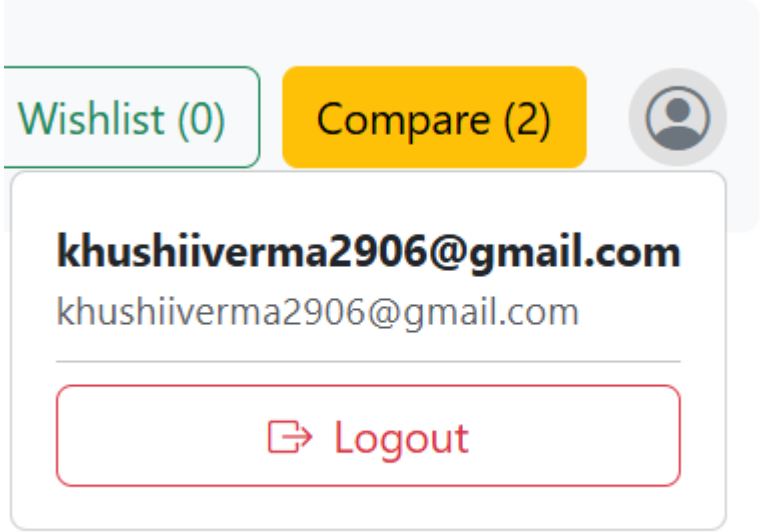

CryptoTracker
Your Wishlist
You have no coins in your wishlist.

Compare:

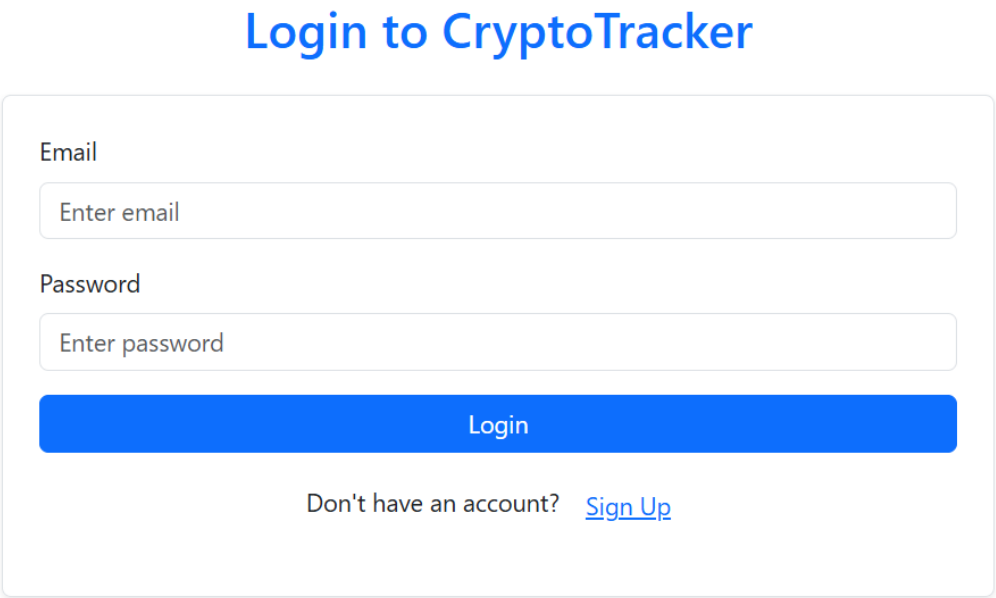

CryptoTracker
Compare Coins
Tether
Price: $1
Market Cap: $1,53,19,63,57,935
24h Change: -0.00%
Remove
Tether
5,00,00,00,00,000
-5,00,00,00,00,000
Price
Market Cap
24h Change (%)
Reset Compare


CryptoTracker
Compare Coins
Tether
Price: $1
Market Cap: $1,53,19,63,57,935
24h Change: -0.00%
Remove
Ethereum
Price: $2,525.58
Market Cap: $3,05,03,37,56,375
24h Change: 0.59%
Remove
Tether
Ethereum
4,00,00,00,00,000
3,00,00,00,00,000
2,00,00,00,00,000
1,00,00,00,00,000
0
-1,00,00,00,00,000
Price
Market Cap
24h Change (%)
Reset Compare

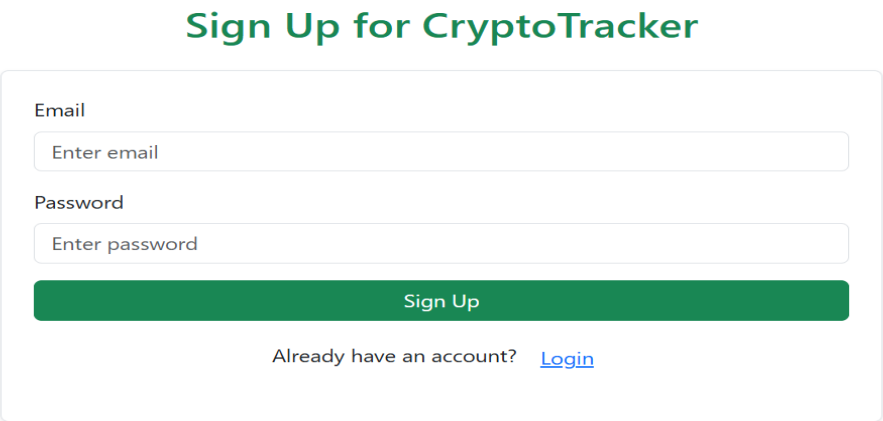
Profile:



Login:



SignUp:


Sign Up for CryptoTracker
Email
Enter email
Password
Enter password
Sign Up
Already have an account?
Login

CHAPTER- 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion:

Crypto-Tracker is an efficient and user-friendly platform that allows users to monitor real-time cryptocurrency prices, compare different coins, and receive important updates via email. It enhances decision-making for both new and experienced investors by providing reliable market data in a secure environment.

6.2 Future Scope:

Future enhancements will focus on adding secure user authentication to personalize experiences, implementing a wishlist and alert system for tracking favorite coins, introducing portfolio management tools to monitor investments and profits, and strengthening security features to protect user data. These improvements will help Crypto-Tracker evolve into a more comprehensive and secure crypto management platform**.**

CHAPTER - 7

REFERENCES

https://reactjs.org/

https://nodejs.org

https://www.w3school.com